Differential Kinematics

The position and orientation of a robot's end effector can be defined by the following relationship

$$Y_i = f_i\left(x_1, x_2, x_3 \cdots, x_j\right), \forall i$$

The differential position and orientation can be calculated by differentiating the position/orientation with respect to each of the joint variables *x*. This gives the following relationship

$$\delta Y_i = \frac{\partial f_i}{\partial x_1}\delta x_1 + \frac{\partial f_i}{\partial x_2}\delta x_2 + \frac{\partial f_i}{\partial x_3}\delta x_3 + \cdots + \frac{\partial f_i}{\partial x_j}\delta x_j, \forall i$$

The Jacobian

This relationship can be represented by the following matrix equation

$$[\delta Y_i] = \left[\frac{\partial f_i}{\partial x_j}\right][\delta x_j]$$

The Jacobian

$$\begin{bmatrix} \delta Y_1 \\ \delta Y_2 \\ \delta Y_3 \\ \vdots \\ \delta Y_i \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \cdots & \frac{\partial f_1}{\partial x_j} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \cdots & \frac{\partial f_2}{\partial x_j} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \cdots & \frac{\partial f_3}{\partial x_j} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_i}{\partial x_1} & \frac{\partial f_i}{\partial x_2} & \frac{\partial f_i}{\partial x_3} & \cdots & \frac{\partial f_i}{\partial x_j} \end{bmatrix} \cdot \begin{bmatrix} \delta x_1 \\ \delta x_2 \\ \delta x_3 \\ \vdots \\ \delta x_j \end{bmatrix}$$

The Robot Jacobian

The corresponding differential equation for a robot manipulator has the following relationship

$$[D]=[J][D_\theta]$$

The Robot Jacobian

Differential End Effector Variables

$$\begin{bmatrix} dx \\ dy \\ dz \\ \delta x \\ \delta y \\ \delta z \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial \theta_1} & \frac{\partial f_1}{\partial \theta_2} & \frac{\partial f_1}{\partial \theta_3} & \frac{\partial f_1}{\partial \theta_4} & \frac{\partial f_1}{\partial \theta_5} & \frac{\partial f_1}{\partial \theta_6} \\ \frac{\partial f_2}{\partial \theta_1} & \frac{\partial f_2}{\partial \theta_2} & \frac{\partial f_2}{\partial \theta_3} & \frac{\partial f_2}{\partial \theta_4} & \frac{\partial f_2}{\partial \theta_5} & \frac{\partial f_2}{\partial \theta_6} \\ \frac{\partial f_3}{\partial \theta_1} & \frac{\partial f_3}{\partial \theta_2} & \frac{\partial f_3}{\partial \theta_3} & \frac{\partial f_3}{\partial \theta_4} & \frac{\partial f_3}{\partial \theta_5} & \frac{\partial f_3}{\partial \theta_6} \\ \frac{\partial f_4}{\partial \theta_1} & \frac{\partial f_4}{\partial \theta_2} & \frac{\partial f_4}{\partial \theta_3} & \frac{\partial f_4}{\partial \theta_4} & \frac{\partial f_4}{\partial \theta_5} & \frac{\partial f_4}{\partial \theta_6} \\ \frac{\partial f_5}{\partial \theta_1} & \frac{\partial f_5}{\partial \theta_2} & \frac{\partial f_5}{\partial \theta_3} & \frac{\partial f_5}{\partial \theta_4} & \frac{\partial f_5}{\partial \theta_5} & \frac{\partial f_5}{\partial \theta_6} \\ \frac{\partial f_6}{\partial \theta_1} & \frac{\partial f_6}{\partial \theta_2} & \frac{\partial f_6}{\partial \theta_3} & \frac{\partial f_6}{\partial \theta_4} & \frac{\partial f_6}{\partial \theta_5} & \frac{\partial f_6}{\partial \theta_6} \end{bmatrix} \cdot \begin{bmatrix} d\theta_1 \\ d\theta_2 \\ d\theta_3 \\ d\theta_4 \\ d\theta_5 \\ d\theta_6 \end{bmatrix}$$

Differential Joint Variables

Calculating The Jacobian

Each Element of the Jacobian is the partial derivative of the corresponding kinematic equation with respect to one of the joint variables.

For example: The element dx indicates that the first kinematic equation must represent movements along the x-axis (i.e. motion relative to p_x). It follows that the derivative of p_x is dx.

The same argument holds for dy and dz.

The Jacobian Relative to the Last Frame

It is simpler to associate the Jacobian Elements to the last moving reference frame (*T6*) rather than to the first. This means that the differential/velocity relationship can be written:

$$\left[{}^{T6}D\right] = \left[{}^{T6}J\right]\left[D_{\theta}\right]$$

This relationship can be calculated with respect to the last reference frame in the following way:

The differential motion relationship can be written as

$$\begin{bmatrix} {}^{T6}dx \\ {}^{T6}dy \\ {}^{T6}dz \\ {}^{T6}\delta x \\ {}^{T6}\delta y \\ {}^{T6}\delta z \end{bmatrix} = \begin{bmatrix} {}^{T6}J_{11} & {}^{T6}J_{12} & \cdots & {}^{T6}J_{16} \\ {}^{T6}J_{21} & {}^{T6}J_{22} & \cdots & {}^{T6}J_{26} \\ {}^{T6}J_{31} & {}^{T6}J_{32} & \cdots & {}^{T6}J_{36} \\ {}^{T6}J_{41} & {}^{T6}J_{42} & \cdots & {}^{T6}J_{46} \\ {}^{T6}J_{51} & {}^{T6}J_{52} & \cdots & {}^{T6}J_{56} \\ {}^{T6}J_{61} & {}^{T6}J_{62} & \cdots & {}^{T6}J_{66} \end{bmatrix} \cdot \begin{bmatrix} d\theta_1 \\ d\theta_2 \\ d\theta_3 \\ d\theta_4 \\ d\theta_5 \\ d\theta_6 \end{bmatrix}$$

The joint transformations (A_i) can be combined and then represented as an *n,o,a,p* matrix

The elements of *n,o,a,p* matrix can then be associated to the Jacobian. This association depends on the type of joint being considered

For a revolute joint the Jacobian column elements are:

$$ {}^{T6}J_{1i} = \left(-n_x p_y + n_y p_x\right) \quad {}^{T6}J_{2i} = \left(-o_x p_y + o_y p_x\right) \quad {}^{T6}J_{3i} = \left(-a_x p_y + a_y p_x\right) $$

$$ {}^{T6}J_{4i} = n_z \qquad {}^{T6}J_{5i} = o_z \qquad {}^{T6}J_{6i} = a_z $$

For a prismatic joint the Jacobian column elements are:

$$ {}^{T6}J_{1i} = n_z \qquad {}^{T6}J_{2i} = o_z \qquad {}^{T6}J_{3i} = a_z $$

$$ {}^{T6}J_{4i} = 0 \qquad {}^{T6}J_{5i} = 0 \qquad {}^{T6}J_{6i} = 0 $$

The associated *n,o,a,p* elements for each column come from the following:

$$ {}^{i-1}T_6 = \prod_{j=i}^{6} A_j $$

The Jacobian and the Differential Operator

The elements of $[\Delta]$ are the differential position and orientation variables for the end effector frame. These variables are used to calculate the Differential Operator:

$$\Delta = \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\delta z & \delta y & 0 \\ \delta z & 1 & -\delta x & 0 \\ -\delta y & \delta x & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Similarly, the elements of $[{}^{T6}\Delta]$ can be used to define the Differential Operator for a moving reference frame:

$$[T^{-1}][\Delta][T] = {}^{T6}\Delta = \begin{bmatrix} 0 & -{}^{T6}\delta z & {}^{T6}\delta y & {}^{T6}dx \\ {}^{T6}\delta z & 0 & -{}^{T6}\delta x & {}^{T6}dy \\ -{}^{T6}\delta y & {}^{T6}\delta x & 0 & {}^{T6}dz \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The Inverse Jacobian

The Inverse Jacobian can be used to calculate the joint differential motions/velocities in order to achieve the desired end effector differential motions/velocities:

$$\left[J^{-1}\right]\left[D\right] = \left[J^{-1}\right]\left[J\right]\left[D_\theta\right]$$
$$\therefore \left[D_\theta\right] = \left[J^{-1}\right]\left[D\right]$$

Alternatively:

$$\left[{}^{T6}J^{-1}\right]\left[{}^{T6}D\right] = \left[{}^{T6}J^{-1}\right]\left[{}^{T6}J\right]\left[D_\theta\right]$$
$$\therefore \left[D_\theta\right] = \left[{}^{T6}J^{-1}\right]\left[{}^{T6}D\right]$$